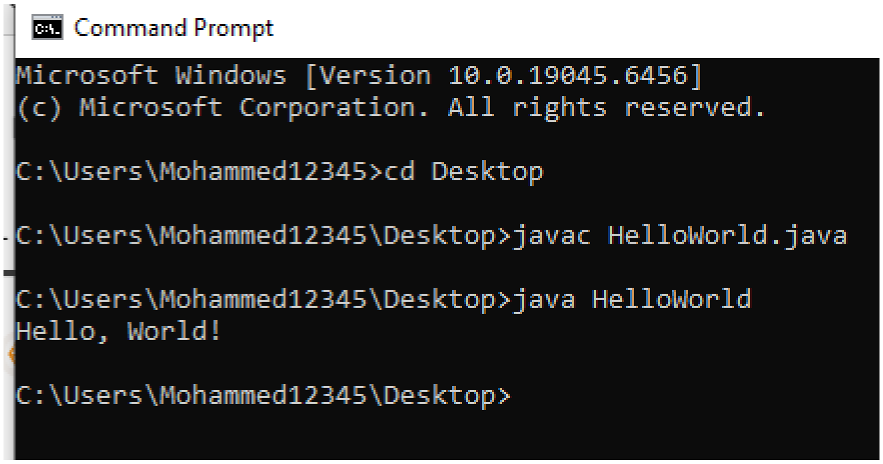
SET-01 (Beginner)

Problem-1:

Print "Hello, World!"

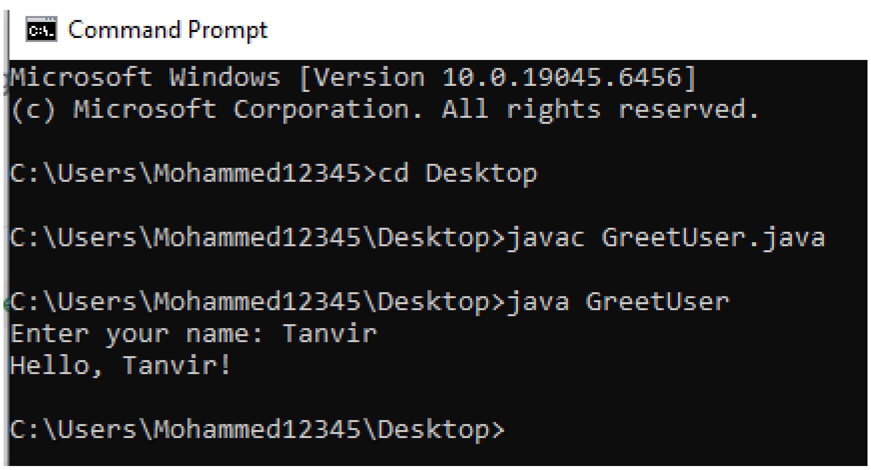


Approach:

I started by creating a simple Java program with a main method. Inside the main method, I used System.out.println() to print “Hello, World!” on the screen. After compiling and running the program, it successfully displayed the output.

Problem-2:

Take the user's name and greet them. Example, Input: "Your Name" and Output: Hello, "Your Name"!



Approach:

I created a Java program to take user input using the Scanner class. First, I asked the user to enter their name. Then I stored the input in a variable and used System.out.println() to display a greeting message like “Hello, Tanvir!”. The program successfully showed the input and greeting output.

SET-01 (Beginner)

Problem-3:

Check if a number is even or odd

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac EvenOdd.java

C:\Users\Mohammed12345\Desktop>java EvenOdd
Enter a number: 7
Odd number

C:\Users\Mohammed12345\Desktop>
```

Approach:

I created a Java program to take a number as input using the Scanner class. Then I used the modulus operator (%) to check if the number is divisible by 2. If the remainder is 0, it prints “Even number”; otherwise, it prints “Odd number” using System.out.println(). The program successfully showed the correct result.

Problem-4:

Find the maximum of 3 numbers: Take 3 numbers from user input, print the largest

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac MaxOfThree.java

C:\Users\Mohammed12345\Desktop>java MaxOfThree
Enter first number: 5
Enter second number: 9
Enter third number: 3
Maximum number is: 9

C:\Users\Mohammed12345\Desktop>
```

Approach:

I created a Java program to take three numbers as input using the Scanner class. Then I used if-else statements to compare the three numbers. I checked which number is greater than the others and stored it as the maximum. Finally, I used System.out.println() to display the largest number. The program successfully showed the correct result.

SET-01 (Beginner)

Problem-5:

Simple Calculator (Add, Subtract, Multiply, Divide). Use if-else or switch to perform operations.

Command Prompt - java Calculator

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac Calculator.java

C:\Users\Mohammed12345\Desktop>java Calculator
Enter first number: 10
Enter second number: 5
Enter operator (+, -, *, /):
```

Approach:

I created a Java program to take two numbers and an operator (+, -, *, /) as input using the Scanner class. Then I used if-else or switch statements to check the operator. Based on the operator, I performed the corresponding calculation (addition, subtraction, multiplication, or division). Finally, I displayed the result using System.out.println(). The program successfully performed the selected operation and showed the output.

Problem-6:

Print numbers from 1 to N. Take N from the user input.

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac PrintNumbers.java

C:\Users\Mohammed12345\Desktop>java PrintNumbers
Enter a number (N): 5
1
2
3
4
5

C:\Users\Mohammed12345\Desktop>_
```

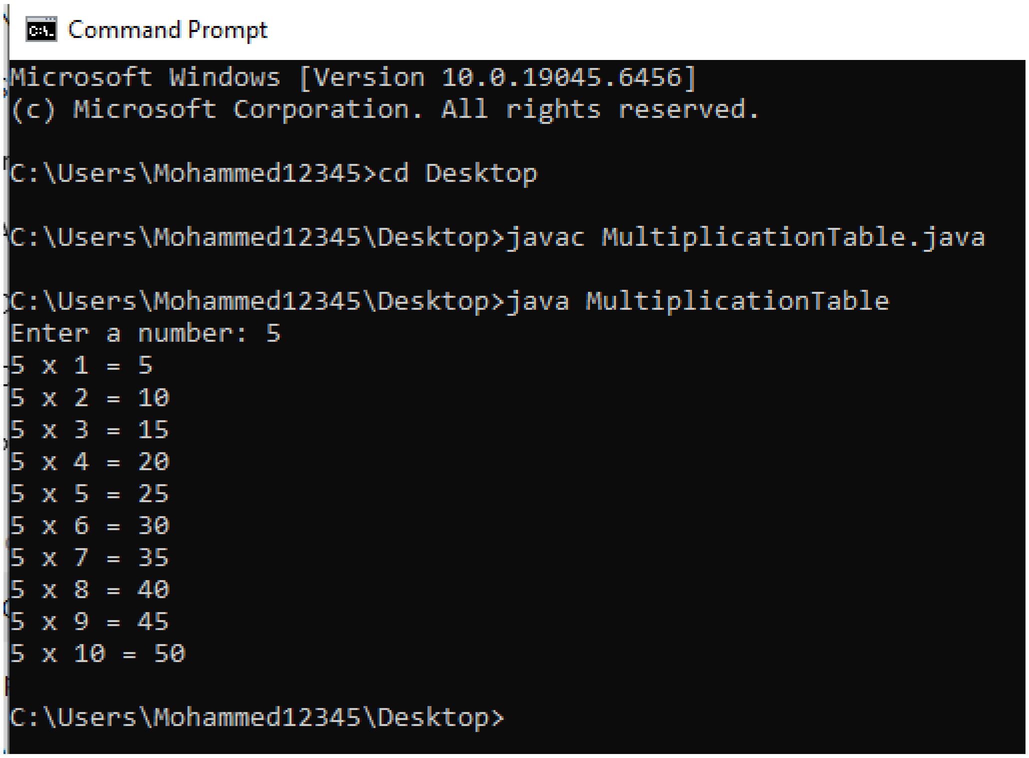
Approach:

I created a Java program to take a number (N) as input using the Scanner class. Then I used a loop (for or while) to print numbers from 1 up to N. In each iteration, I displayed the number using System.out.println(). The loop continued until it reached N. The program successfully showed all numbers from 1 to N.

SET-01 (Beginner)

Problem-7:

Print the multiplication table of a number



```
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac MultiplicationTable.java

C:\Users\Mohammed12345\Desktop>java MultiplicationTable
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

C:\Users\Mohammed12345\Desktop>
```

Approach:

I created a Java program to take a number as input using the Scanner class. Then I used a loop (for loop) to iterate from 1 to 10. In each iteration, I multiplied the input number with the loop variable and displayed the result in the format "5 x 1 = 5" using System.out.println(). The program successfully printed the multiplication table.

SET-01 (Beginner)

Problem-8:

Calculate the factorial of a number. Example: 515×4×3×2×1=120

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac Factorial.java

C:\Users\Mohammed12345\Desktop>java Factorial
Enter a number: 5
Factorial = 120

C:\Users\Mohammed12345\Desktop>
```

Approach:

I created a Java program to take a number as input using the Scanner class. Then I used a loop (for loop) to multiply numbers from 1 up to the given number. I stored the result in a variable (factorial). In each step, the value was updated by multiplying the current number. Finally, I displayed the result using System.out.println(). The program successfully showed the factorial.

Problem-9:

Sum of all numbers from I to N

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac SumFromOneToN.java

C:\Users\Mohammed12345\Desktop>java SumFromOneToN
Enter a number (N): 5
Sum = 15

C:\Users\Mohammed12345\Desktop>_
```

Approach:

I created a Java program to take a number (N) as input using the Scanner class. Then I initialized a variable sum to 0 and used a loop (for loop) to add numbers from 1 to N. In each iteration, I added the current number to sum. Finally, I displayed the total using System.out.println(). The program successfully showed the sum.

SET - 02(Intermediate)

Problem-1:

Store and print 5 numbers in an array

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac ArrayExample.java

C:\Users\Mohammed12345\Desktop>java ArrayExample
Enter 5 numbers:
10
20
30
40
50
You entered:
10
20
30
40
50

C:\Users\Mohammed12345\Desktop>
```

Approach:

I created a Java program to store 5 numbers using an array. I used the Scanner class to take input from the user and stored each number in the array using a loop. Then I used another loop to print all the stored numbers using System.out.println(). The program successfully displayed all the entered numbers.

SET - 02(Intermediate)

Problem-2:

Find the largest number in an array

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac LargestInArray.java

C:\Users\Mohammed12345\Desktop>java LargestInArray
Enter 5 numbers:
10
25
7
40
15
Largest number is: 40

C:\Users\Mohammed12345\Desktop>
```

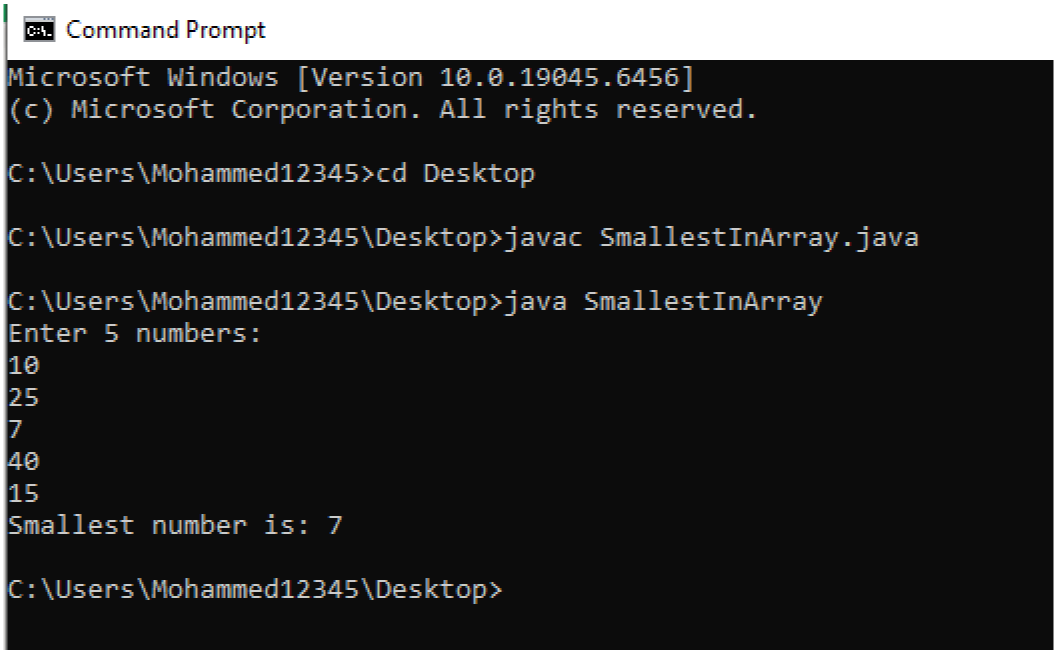
Approach:

I created a Java program to take 5 numbers as input using the Scanner class and stored them in an array. Then I assumed the first element as the largest and used a loop to compare it with the remaining elements. If a bigger number was found, I updated the largest value. Finally, I displayed the result using System.out.println(). The program successfully showed the largest number.

SET - 02(Intermediate)

Problem-3:

Find the smallest number in an array



Approach:

I created a Java program to take 5 numbers as input using the Scanner class and stored them in an array. Then I assumed the first element as the smallest and used a loop to compare it with the remaining elements. If a smaller number was found, I updated the smallest value. Finally, I displayed the result using System.out.println(). The program successfully showed the smallest number.

SET - 02(Intermediate)

Problem-4:

Calculate the average of an array

```
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac AverageArray.java

C:\Users\Mohammed12345\Desktop>java AverageArray
Enter 5 numbers:
10
20
30
40
50
Average = 30.0

C:\Users\Mohammed12345\Desktop>
```

Approach:

I created a Java program to take 5 numbers as input using the Scanner class and stored them in an array. Then I initialized a variable sum to 0 and used a loop to add all the elements of the array. After calculating the total, I divided the sum by the number of elements to find the average. Finally, I displayed the result using System.out.println(). The program successfully showed the average.

SET - 02(Intermediate)

Problem-5:

Count even and odd numbers in an array

```
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac CountEvenOdd.java

C:\Users\Mohammed12345\Desktop>java CountEvenOdd
Enter 5 numbers:
1
2
3
4
5
Even numbers: 2
Odd numbers: 3

C:\Users\Mohammed12345\Desktop>
```

Approach:

I created a Java program to take 5 numbers as input using the Scanner class and stored them in an array. Then I initialized two variables to count even and odd numbers. I used a loop to check each element of the array using the modulus operator (%). If a number is divisible by 2, I increased the even count; otherwise, I increased the odd count. Finally, I displayed the results using System.out.println(). The program successfully showed the count of even and odd numbers.

SET - 02(Intermediate)

Problem-6:

Search for an element in an array

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac SearchElement.java

C:\Users\Mohammed12345\Desktop>java SearchElement
Enter 5 numbers:
10
20
30
40
50
Enter number to search: 30
Element found

C:\Users\Mohammed12345\Desktop>
```

Approach:

I created a Java program to take 5 numbers as input using the Scanner class and stored them in an array. Then I took another input as the number to search. I used a loop to check each element of the array and compared it with the search value. If a match was found, I displayed “Element found”; otherwise, I showed “Element not found” using System.out.println(). The program successfully searched the element.

SET - 02(Intermediate)

Problem-7:

Sort an array (Ascending Order)

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac SortArray.java

C:\Users\Mohammed12345\Desktop>java SortArray
Enter 5 numbers:
50
20
40
10
30
Sorted array (Ascending):
10
20
30
40
50

C:\Users\Mohammed12345\Desktop>
```

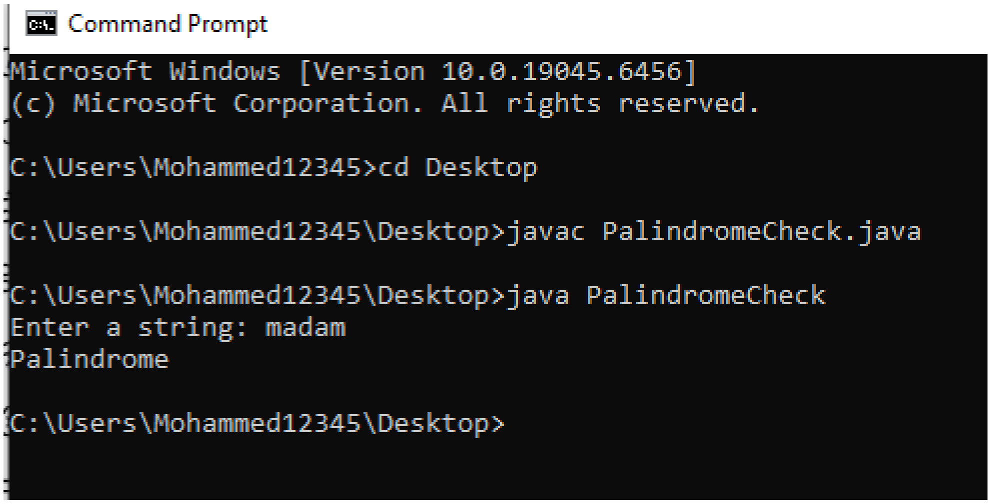
Approach:

I created a Java program to take 5 numbers as input using the Scanner class and stored them in an array. Then I used a sorting technique (like nested loops) to compare each element with others. If a smaller element was found, I swapped the values. After completing the sorting process, I used System.out.println() to display the sorted array in ascending order. The program successfully showed the sorted result.

SET - 02(Intermediate)

Problem-8:

Check if a string is a palindrome



```
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac PalindromeCheck.java

C:\Users\Mohammed12345\Desktop>java PalindromeCheck
Enter a string: madam
Palindrome

C:\Users\Mohammed12345\Desktop>
```

Approach:

I created a Java program to take a string as input using the Scanner class. Then I reversed the string using a loop or built-in method. After that, I compared the original string with the reversed string. If both are equal, I displayed "Palindrome"; otherwise, I showed "Not Palindrome" using System.out.println(). The program successfully checked the result.

SET - 02(Intermediate)

Problem-9:

Count vowels and consonants in a string

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac VowelConsonantCount.java

C:\Users\Mohammed12345\Desktop>java VowelConsonantCount
Enter a string: Hello world
Vowels: 3
Consonants: 7

C:\Users\Mohammed12345\Desktop>
```

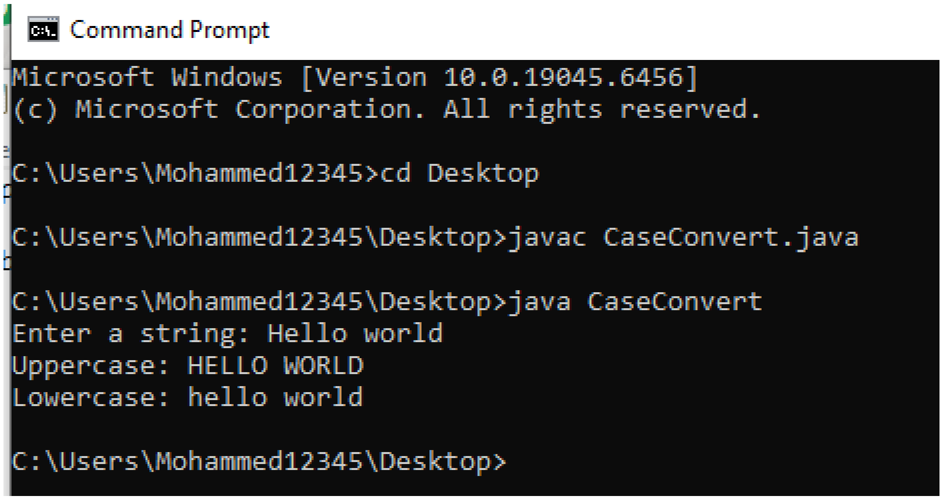
Approach:

I created a Java program to take a string as input using the Scanner class. Then I initialized two variables to count vowels and consonants. I used a loop to go through each character of the string. If the character is a vowel (a, e, i, o, u), I increased the vowel count; if it is an alphabet but not a vowel, I increased the consonant count. Finally, I displayed the results using System.out.println(). The program successfully showed the counts.

SET - 02(Intermediate)

Problem-10:

Convert a string to uppercase and lowercase



```
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac CaseConvert.java

C:\Users\Mohammed12345\Desktop>java CaseConvert
Enter a string: Hello world
Uppercase: HELLO WORLD
Lowercase: hello world

C:\Users\Mohammed12345\Desktop>
```

Approach:

I created a Java program to take a string as input using the Scanner class. Then I used built-in string methods toUpperCase() and toLowerCase() to convert the string into uppercase and lowercase. Finally, I displayed both results using System.out.println(). The program successfully showed the converted strings.

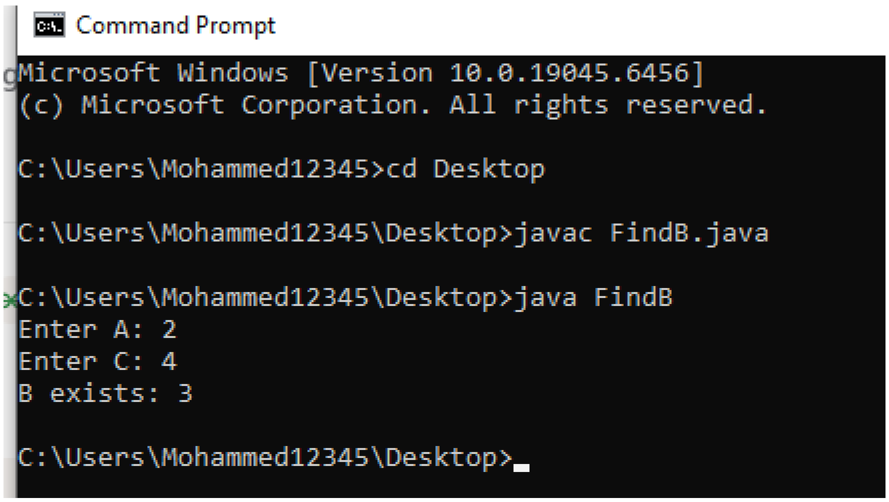
SET - 03 (Descriptive Problems)

Problem-1:

You are given 2 integers A and C. You need to find if there exists any integer B that meets the following condition

-B must be an integer.

-B is the average of A and C



Approach:

I created a Java program to take two integers A and C as input using the Scanner class. Then I calculated the average using the formula (A + C) / 2. I checked whether the result is an integer by ensuring (A + C) is divisible by 2. If it is divisible, I stored the value as B and displayed it using System.out.println(). Otherwise, I showed that B does not exist. The program successfully found whether B exists or not.

SET - 03 (Descriptive Problems)

Problem-2:

Sometimes, some words like "localization" or "internationalization" are so long that writing them many times in one text is quite tiresome.

Let's consider a word too long, if its length is strictly more than 10 characters. All too long words should be replaced with a special abbreviation.

This abbreviation is made like this: we write the first and last letters of a word, and between them we write the number of letters between them. That number is in decimal and doesn't contain any leading zeros.

Thus, "localization" will be spelt as "110n", and "internationalization" will be spelt as "i18n".

You are suggested to automate the process of changing words with abbreviations. At that, all too long words should be replaced by the abbreviation and the words that are not too long should not undergo any changes.

```
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac WordAbbreviation.java

C:\Users\Mohammed12345\Desktop>java WordAbbreviation
Enter number of words: 3
localization
l10n
internationalization
i18n
hello
hello

C:\Users\Mohammed12345\Desktop>
```

Approach:

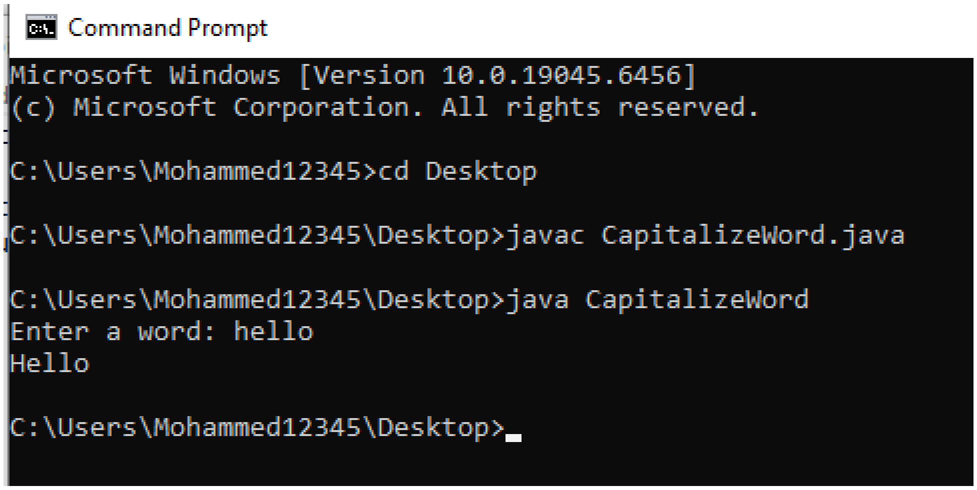
I created a Java program to take the number of words as input using the Scanner class. Then I used a loop to read each word one by one. For each word, I checked its length. If the length is greater than 10, I created an abbreviation by taking the first letter, the number of middle characters (length - 2), and the last letter. Otherwise, I printed the word as it is. Finally, I displayed the result using System.out.println(). The program successfully converted long words into abbreviations.

SET - 03 (Descriptive Problems)

Problem-3:

Capitalization is writing a word with its first letter as a capital letter. Your task is to capitalize the given word.

Note that during capitalization, all the letters except the first one remain unchanged.



```
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac CapitalizeWord.java

C:\Users\Mohammed12345\Desktop>java CapitalizeWord
Enter a word: hello
Hello

C:\Users\Mohammed12345\Desktop>
```

Approach:

I created a Java program to take a word as input using the Scanner class. Then I converted the first character of the word to uppercase using Character.toUpperCase(). After that, I kept the remaining part of the word unchanged and combined both parts. Finally, I displayed the result using System.out.println(). The program successfully showed the capitalized word.

SET - 03 (Descriptive Problems)

Problem-4:

You are given three digits, a, b, and c. Two of them are equal, but the third one is different from the other two.

Find the value that occurs exactly once.

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac UniqueNumber.java

C:\Users\Mohammed12345\Desktop>java UniqueNumber
Enter a: 5
Enter b: 5
Enter c: 9
Unique number: 9

C:\Users\Mohammed12345\Desktop>
```

Approach:

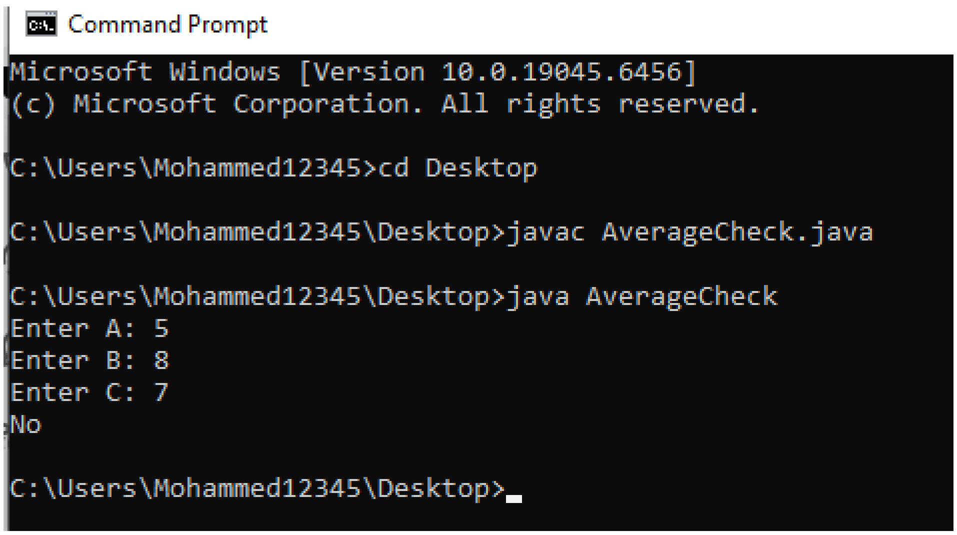
I created a Java program to take three integers a, b, and c as input using the Scanner class. Since two numbers are equal and one is different, I used if-else conditions to compare the values. If a equals b, then c is the unique number; if a equals c, then b is the unique number; otherwise, a is the unique number. Finally, I displayed the result using System.out.println(). The program successfully found the unique number.

SET - 03 (Descriptive Problems)

Problem-5:

You are given 3 numbers A, B, and C. Determine whether the average of A and B is strictly greater than C or not?

NOTE: Average of A and B is defined as (A+B)/2. For example, the average of 5 and 9 is 7, and the average of 5 and 8 is 6.5.



Approach:

I created a Java program to take three numbers A, B, and C as input using the Scanner class. Then I calculated the average of A and B using the formula (A + B) / 2. After that, I compared the average with C. If the average is strictly greater than C, I printed “Yes”; otherwise, I printed “No” using System.out.println(). The program successfully showed the correct result.

SET - 03 (Descriptive Problems)

Problem-6:

A new TV streaming service, IIUC-TV, was recently launched in IIUC Hall. A group of N friends in IIUC wants to buy IIUC-TV subscriptions. We know that 6 people can share a single IIUC-TV subscription, IIUC. Also, the cost of one IIUC-TV subscription is X Taka. Determine the minimum total cost that the group of N friends will incur to use IIUC-TV, IIUC

```
Command Prompt
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac SubscriptionCost.java

C:\Users\Mohammed12345\Desktop>java SubscriptionCost
Enter number of friends (N): 10
Enter cost per subscription (X): 100
Minimum total cost: 200

C:\Users\Mohammed12345\Desktop>
```

Approach:

I created a Java program to take the number of friends (N) and the cost per subscription (X) as input using the Scanner class. Since one subscription can be shared by 6 people, I calculated the required number of subscriptions by dividing N by 6 and rounding up (if there is any remainder, one extra subscription is needed). Then I multiplied the total number of subscriptions by X to get the minimum total cost. Finally, I displayed the result using System.out.println(). The program successfully showed the minimum cost.

SET - 03 (Descriptive Problems)

Problem-7:

A student has to submit 3 assignments before 10 pm, and he starts to do the assignments at X pm. Each assignment takes him 1 hour to complete. Can you tell whether he'll he able to complete all assignments on time?

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac AssignmentCheck.java

C:\Users\Mohammed12345\Desktop>java AssignmentCheck
Enter starting time (X pm): 7
Yes

C:\Users\Mohammed12345\Desktop>
```

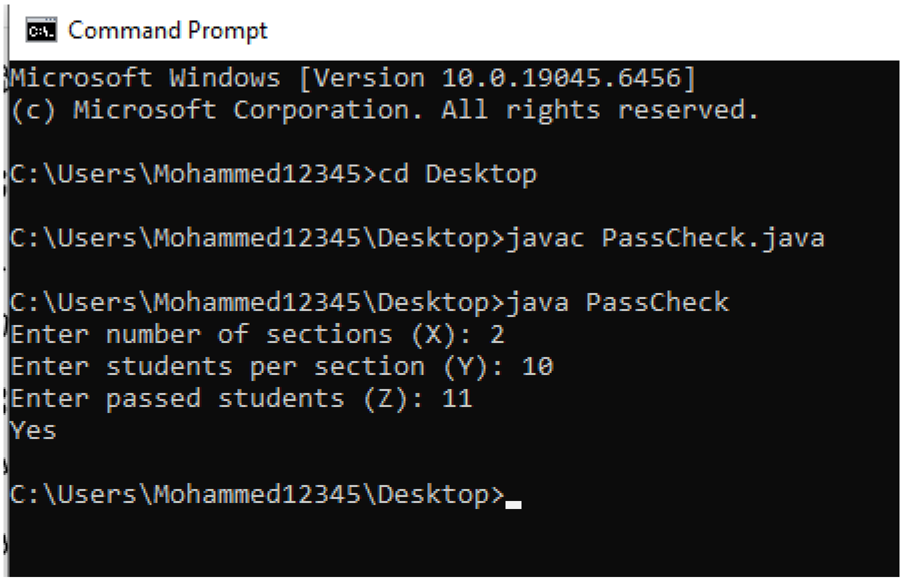
Approach:

I created a Java program to take the starting time (X) as input using the Scanner class. Since each assignment takes 1 hour and there are 3 assignments, the total time needed is 3 hours. I added 3 to the starting time (X + 3) and checked if it is less than or equal to 10 pm. If it is, I printed “Yes”; otherwise, I printed “No” using System.out.println(). The program successfully determined whether the student can finish on time.

SET - 03 (Descriptive Problems)

Problem-8:

In IIUC, there are X sections, and each section has Y students. The semester-end results are in, and a total of Z students passed the exams. Assuming that all students appeared for the exams, find whether the number of students who passed was strictly greater than 50%.



Approach:

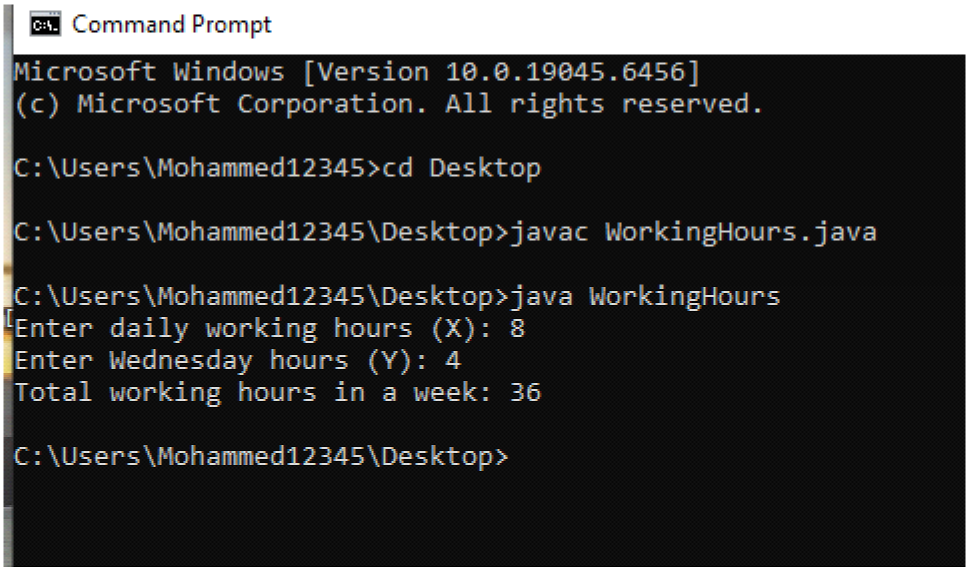
I created a Java program to take X (sections), Y (students per section), and Z (passed students) as input using the Scanner class. Then I calculated the total number of students by multiplying X and Y (total = X × Y). After that, I checked if the number of passed students (Z) is strictly greater than half of the total students (total / 2). If it is greater, I printed “Yes”; otherwise, I printed “No” using System.out.println(). The program successfully determined the result.

SET - 03 (Descriptive Problems)

Problem-9:

Recently You joined a new company. At this company, employees must work X hours each day from Saturday to Wednesday. Also, in this company, Wednesday is called Chill Day, employees only have to work for Y hours (Y<X) on Wednesday.

Determine the total number of working hours in one week.



Approach:

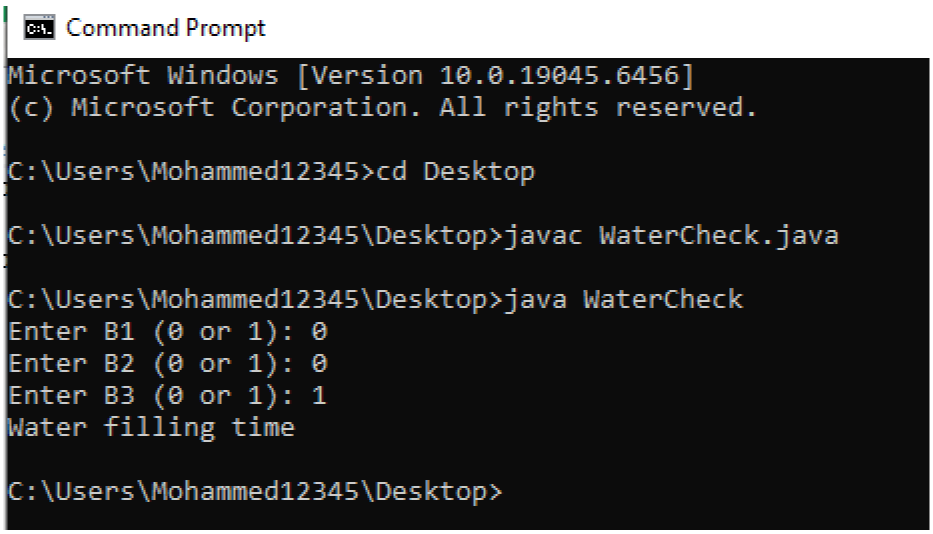
I created a Java program to take daily working hours (X) and Wednesday hours (Y) as input using the Scanner class. Employees work full X hours for 4 days (Saturday to Tuesday) and Y hours on Wednesday. So, I calculated total working hours as $(4 \times X) + Y$. Finally, I displayed the result using System.out.println(). The program successfully showed the total working hours in a week.

SET - 03 (Descriptive Problems)

Problem-10:

An IIUCian has three water bottles. At any point, if at least two of them are empty, he will fill them up. But if at most one bottle is empty, he will wait and not fill them up now. You are given three integers: B1, B2, and B3. If B1-1, then the first bottle is full. If B1-0, then the first bottle is empty. Similarly, B2 denotes whether the second bottle is

full or empty, and B3 denotes it for the third bottle. Output "Water filling time", if he has to fill the bottles now. If not, output "Not now".



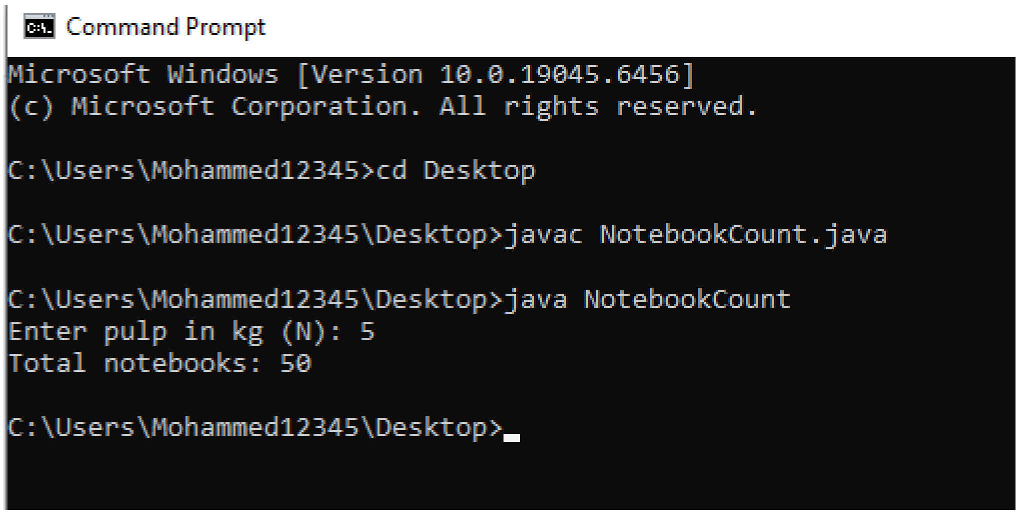
Approach:

I created a Java program to take three inputs B1, B2, and B3 using the Scanner class, where 0 means empty and 1 means full. Then I counted how many bottles are empty (value = 0). If the number of empty bottles is 2 or more, I printed “Water filling time”; otherwise, I printed “Not now” using System.out.println(). The program successfully determined the correct output.

SET - 03 (Descriptive Problems)

Problem-11:

You know that 1 kg of pulp can be used to make 1000 pages and 1 notebook consists of 100 pages. Suppose a notebook factory receives N kg of pulp. How many notebooks can be made from that?



Approach:

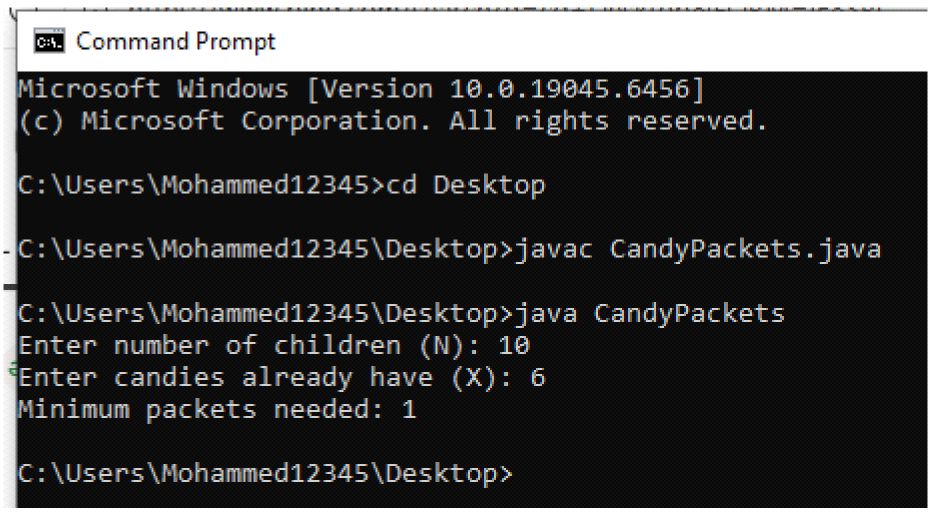
I created a Java program to take the amount of pulp (N kg) as input using the Scanner class. Since 1 kg pulp makes 1000 pages, I calculated total pages as N × 1000. Then, since each notebook contains 100 pages, I divided the total pages by 100 to find the number of notebooks. Finally, I displayed the result using System.out.println(). The program successfully showed the total number of notebooks.

SET - 03 (Descriptive Problems)

Problem-12:

There are N children, and an IIUCian wants to give each of them 1 candy. He already has X candies. To buy the rest, he visits a candy shop. In the shop, packets containing exactly 4 candies are available.

Determine the minimum number of candy packets he must buy so that he can give 1 candy packet to each of the N children.



Approach:

I created a Java program to take the number of children (N) and candies already available (X) using the Scanner class. First, I calculated how many more candies are needed by subtracting X from N (needed = N – X). If needed is less than or equal to 0, no packets are required. Otherwise, since each packet contains 4 candies, I divided the needed candies by 4 and rounded up to get the minimum number of packets. Finally, I displayed the result using System.out.println(). The program successfully showed the minimum packets needed.

SET - 03 (Descriptive Problems)

Problem-13:

An IIUCian has reached the finals of the Annual Inter-university Declamation contest. For the finals, students were asked to prepare 10 topics. However, he was only able to prepare three topics, numbered A, B, and C; he is totally blank about the other topics. This means he can only win the contest if he gets to speak on topics A, B, or C. On the contest day, he gets topic X.

Determine whether he has any chances of winning the competition. Print "Yes" if it is possible to win the contest, else print "No".

Command Prompt

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mohammed12345>cd Desktop

C:\Users\Mohammed12345\Desktop>javac ContestChance.java

C:\Users\Mohammed12345\Desktop>java ContestChance
Enter topic A: 2
Enter topic B: 4
Enter topic C: 6
Enter given topic X: 4
Yes

C:\Users\Mohammed12345\Desktop>
```

Approach:

I created a Java program to take topics A, B, C, and X as input using the Scanner class. Then I used if-else conditions to check whether X matches any of the prepared topics (A, B, or C). If X is equal to any one of them, I printed “Yes”; otherwise, I printed “No” using System.out.println(). The program successfully determined whether he has a chance to win.

SET - 03 (Descriptive Problems)

Problem-14:

There are 4 companies in the markets of Chittagong, A, B, C, and D. This year,

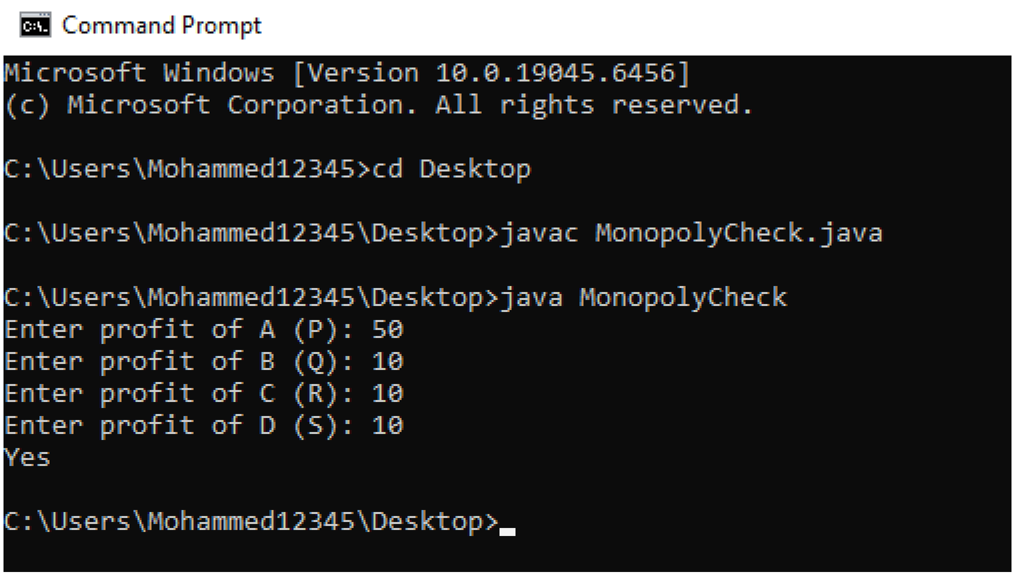
a. Company A made a profit of P lakh,

b. Company B made a profit of Q lakh,

c. Company C made a profit of R lakh,

d. Company D made a profit of S lakh.

There is said to be a monopoly in the market if the profit made by one company is strictly greater than the sum of profits made by all other companies. Determine whether there is a monopoly in the market.



Approach:

I created a Java program to take the profits of four companies P, Q, R, and S as input using the Scanner class. Then I checked each company's profit one by one to see if it is strictly greater than the sum of the other three companies. If any one company satisfies this condition, I printed "Yes"; otherwise, I printed "No" using System.out.println(). The program successfully determined whether there is a monopoly in the market.

SET - 03 (Descriptive Problems)

Problem-15:

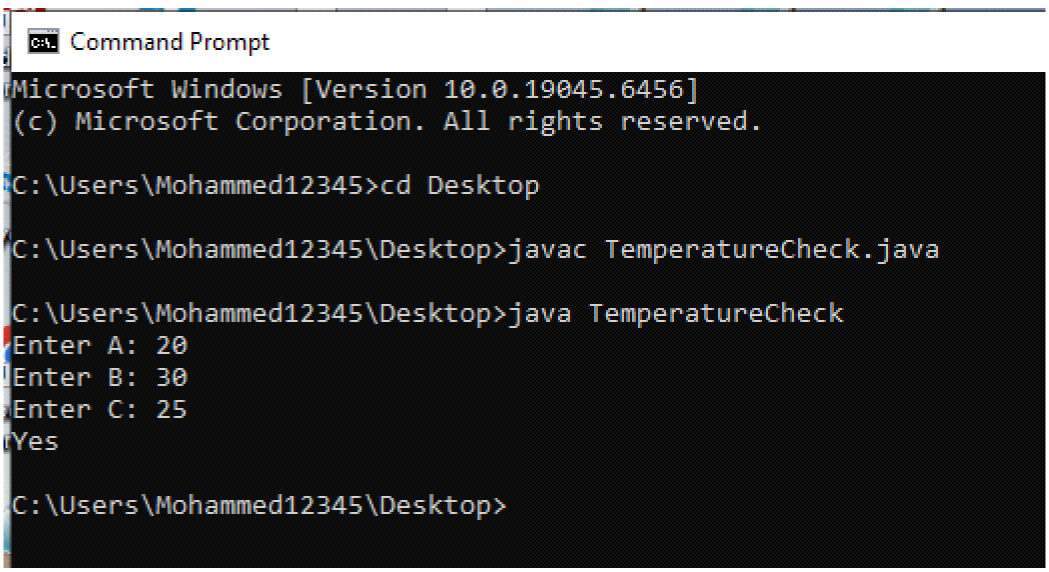
There are three people sitting in a room: Dean Sir, Chairman Sir, and Coordinator Sir. They need to decide on the temperature to set on the air conditioner. Everyone has a demand each:

a.Alice wants the temperature to be at least A degrees.

b. Bob wants the temperature to be at most B degrees.

c. Charlie wants the temperature to be at least C degrees.

Can they all agree on some temperature, or not?



Approach:

I created a Java program to take A, B, and C as input using the Scanner class. Since Alice wants at least A degrees and Charlie wants at least C degrees, the minimum required temperature is the maximum of A and C. Bob wants at most B degrees, so the temperature must be less than or equal to B. I checked if max(A, C) $\leq$ B. If true, I printed "Yes"; otherwise, I printed "No" using System.out.println(). The program successfully determined whether they can agree.